Session 05 (AIML) - Assignment Questions

Q1. (String Indexing)
Write a program that takes a string input from the user.
Print:

- The first character
- The last character (using negative index)
- The 3rd character from the start
- The 2nd character from the end

Add comments explaining positive and negative indexing.

Q2. (String Slicing)
Take a string input from the user.
Using slicing, print:
a) The first 5 characters
b) The last 4 characters
c) The string in reverse order
d) Every 2nd character of the string

Include comments explaining the slicing syntax [start:end:step].

Q3. (String Membership)
Write a program that:

- Takes a main string and a substring as input from the user.
- Checks whether the substring is present in the main string using in and not in.
- Prints appropriate messages (e.g., "Substring found!" or "Substring not found!").

Note that it is case-sensitive.

Q4. (len() Function)
Write a program that:

- Takes a string as input.
- Prints the length of the string using len().
- Prints the last character using len() (without using negative index).
- Prints the middle character if the length is odd.

Add comments on common mistakes with len().

Q5. (range() - Basic Forms)
Write a program using for loop and range() to:
a) Print numbers from 0 to 10.
b) Print numbers from 5 to 15.
c) Print odd numbers from 1 to 21.

Explain the three forms of range() in comments.

Q6. (range() with Negative Step)
Use range() to print:

- Numbers from 20 down to 10 (decreasing).
- Numbers from 15 down to 1 in steps of 2.

Print each sequence on a new line with proper messages.

Q7. (Combined: Strings + range())
Write a program that takes a string from the user.
Using a for loop and range() with len(), print:

- Each character of the string one by one (with its index).
- The string in reverse order using negative step in range().

Q8. (Membership with range())
Write a program that:

- Takes a number as input from the user.
- Checks if the number is present in range(1, 51) and in range(10, 100, 5).

- Prints clear messages for each check.

Q9. (Slicing + range())
Write a program that prints the following using range() and string slicing concepts where applicable:

- First 10 even numbers (2 to 20).
- Numbers from 1 to 30 in steps of 3.
- A string "PythonProgramming" sliced to show "Python", "Programming", and the reverse.

Q10. (Mini Project - Combined)
Create a **String Analyzer** program:

- Take a string input from the user.
- Using functions or directly:
    1. Print length of the string.
    2. Print first half and second half using slicing.
    3. Check if the word "python" (case insensitive — you may convert) is present.
    4. Use range() to print all characters with their positive and negative indices.
    5. Print the string in reverse.

Add proper comments and use meaningful variable names.